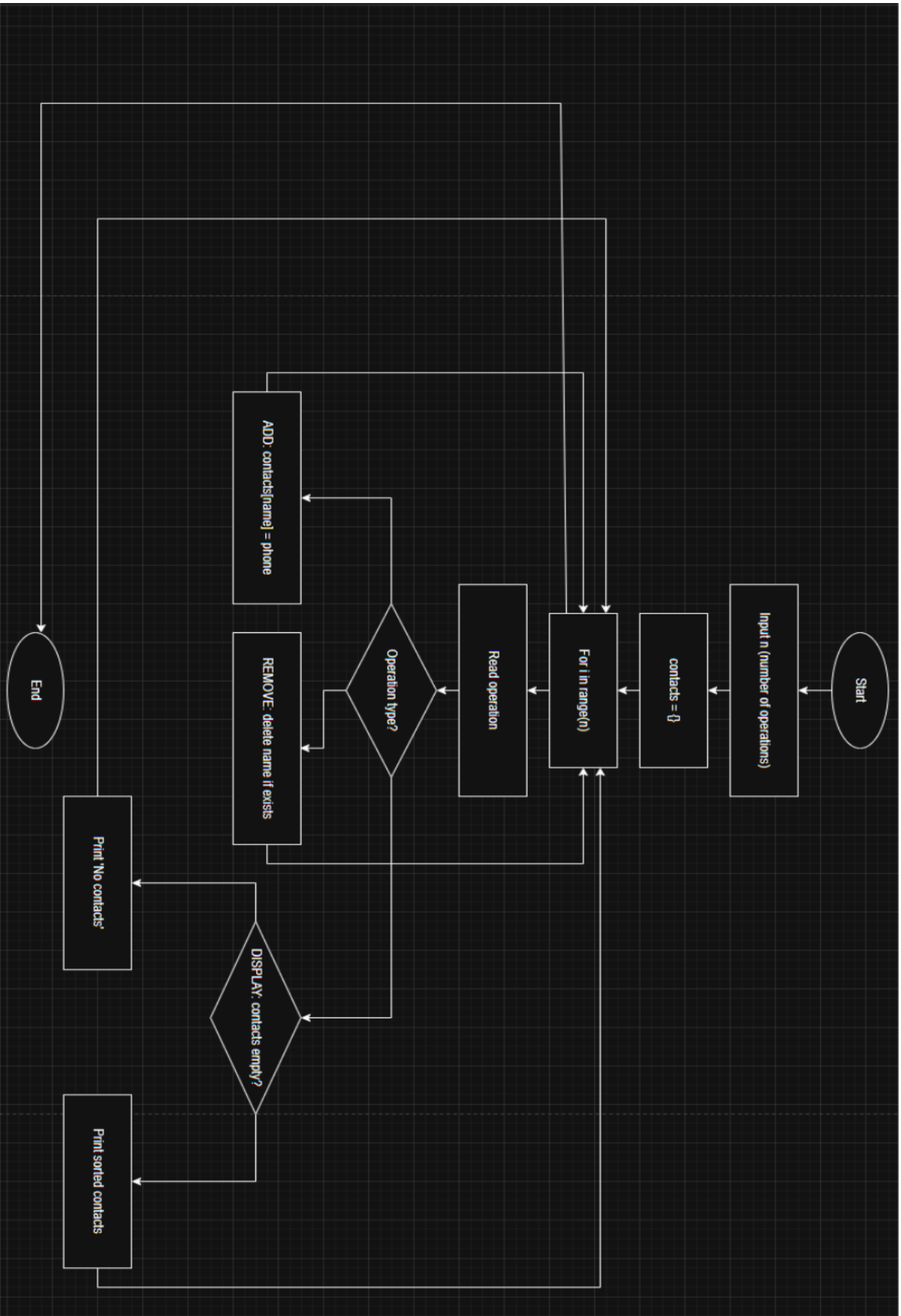


12.1.1 Phone Number Database Management

ALGORITHM

1. Start
2. Input an integer n (number of operations)
3. Initialize an empty dictionary: contacts $\square \{ \}$
4. Repeat the following steps for i = 1 to n:
 - 4.1 Read the operation (split into parts)
 - 4.2 If operation is "ADD":
Extract name and phone_number
If name already exists in contacts:
Update the phone number
Else:
Add new entry
contacts[name] $\square$ phone_number
 - 4.3 Else if operation is "REMOVE":
Extract name
If name exists in contacts:
Remove it
Else:
Ignore operation
 - 4.4 Else if operation is "DISPLAY":
If contacts is empty:
Print "No contacts"
Else:
Sort all names in alphabetical order
For each name in sorted list:
Print name : phone_number
5. End loop
6. End

FLOWCHART



12.1.1: Phone Number Database Management

01:39

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.

Input Format:

1. First line contains an integer n representing the number of operations.
2. Each of the next n lines contains a single operation in one of the following formats:
 - "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
 - "REMOVE <name>" - name is a string.
 - "DISPLAY" - no additional parameters.

Output Format:

- For display operation, print all contacts in alphabetical order by name in the format:

<name>: <phone_number>

- If the database is empty during display operation, print: "No contacts".
- Do not print anything for add and remove operations.

Constraints:

- Name contains only alphabetic characters (no spaces). And it is case sensitive.

Sample Test Cases

+

phoneDat...

```
1 n = int(input())
2 ph={}
3
4 for i in range(n):
5     cmd = input().split()
6
7     if cmd[0] == "ADD":
8         ph[cmd[1]] = cmd[2]
9
10    elif cmd[0] == "REMOVE":
11        if cmd[1] in ph:
12            del ph[cmd[1]]
13
14    elif cmd[0] == "DISPLAY":
15        if not ph:
16            print("No contacts")
17        else:
18            for name in sorted(ph):
19                print(f"{name}: {ph[name]}")
```

Average time
0.006 s
6.38 ms

Maximum time
0.010 s
10.00 ms

5 out of 5 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output

Actual output

```
4 ADD Alice 9876543210 ADD Alice 9876543210
4 ADD Bob 8765432109 ADD Bob 8765432109
4 ADD Charlie 7777777777 ADD Charlie 7777777777
DISPLAY DISPLAY
Alice: 9876543210- Alice: 9876543210-
```

Terminal

Test cases

Submit

PROGRAM

< Prev Reset Submit Next >

